

EE 66
Fall 2026

Signals, Dynamics, and Information

Discussion 5A

1. How Many Frequencies Are There?

For an integer k , define the vector $\psi_k \in \mathbb{C}^p$ by

$$\psi_k[n] = e^{i\frac{2\pi}{p}kn}, \quad n = 0, 1, \dots, p-1.$$

Note that k is allowed to be *any* integer here, not just $0, 1, \dots, p-1$.

- (a) Show that $\psi_{k+p} = \psi_k$. How many distinct vectors are there in the set $\{\psi_k : k \in \mathbb{Z}\}$?

Answer: We compare the two vectors entry by entry. For each $n = 0, 1, \dots, p-1$,

$$\psi_{k+p}[n] = e^{i\frac{2\pi}{p}(k+p)n} = e^{i\frac{2\pi}{p}kn} e^{i\frac{2\pi}{p}pn} = e^{i\frac{2\pi}{p}kn} e^{i2\pi n}.$$

Since n is an integer, $e^{i2\pi n} = 1$, so

$$\psi_{k+p}[n] = e^{i\frac{2\pi}{p}kn} = \psi_k[n]$$

for every n , which means $\psi_{k+p} = \psi_k$ as vectors.

Thus, increasing the index k by p gets us back exactly the vector we started with. So even though k ranges over all the integers, there are only p distinct vectors in the set!

- (b) Show that $\psi_{p-k} = \psi_k^*$, where ψ_k^* denotes the entrywise complex conjugate of ψ_k .

Answer: Using part (a) with k replaced by $-k$, we have $\psi_{p-k} = \psi_{-k}$. So it is enough to show $\psi_{-k} = \psi_k^*$. For each n ,

$$\psi_{-k}[n] = e^{i\frac{2\pi}{p}(-k)n} = e^{-i\frac{2\pi}{p}kn} = \left(e^{i\frac{2\pi}{p}kn}\right)^* = \psi_k^*[n],$$

where the third equality holds because conjugating $e^{i\theta} = \cos \theta + i \sin \theta$ flips the sign of the imaginary part, giving $\cos \theta - i \sin \theta = e^{-i\theta}$.

Therefore $\psi_{p-k} = \psi_{-k} = \psi_k^*$.

- (c) Let $p = 6$. What are ψ_0 , ψ_3 , and ψ_5 ? Interpreted as a 6-periodic DT signal, which has the highest frequency?

Answer: With $p = 6$ we have $\psi_k[n] = e^{i\frac{\pi}{3}kn}$.

For $k = 0$, every entry is $e^0 = 1$. For $k = 3$, we get $\psi_3[n] = e^{i\pi n} = (-1)^n$. For $k = 5$, part (b) tells us $\psi_5 = \psi_{6-1} = \psi_1^*$, so we can evaluate ψ_1 and conjugate. Using Euler's formula on $\psi_1[n] = e^{i\frac{\pi}{3}n}$ and conjugating:

$$\psi_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad \psi_3 = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}, \quad \psi_5 = \begin{bmatrix} 1 \\ \frac{1}{2} - i\frac{\sqrt{3}}{2} \\ -\frac{1}{2} - i\frac{\sqrt{3}}{2} \\ -1 \\ -\frac{1}{2} + i\frac{\sqrt{3}}{2} \\ \frac{1}{2} + i\frac{\sqrt{3}}{2} \end{bmatrix}.$$

(You can also get the entries of ψ_5 directly.)

ψ_3 has the highest frequency. It is the alternating signal $1, -1, 1, -1, \dots$, which flips sign at every single sample — this is as fast as a discrete-time signal can possibly oscillate. ψ_0 is constant, so its frequency is 0.

It is tempting to guess ψ_5 , since 5 is the largest index. But by part (b), $\psi_5 = \psi_1^*$, so ψ_5 oscillates at exactly the same rate as ψ_1 : it just goes around the unit circle in the opposite direction. It takes all 6 samples to complete one full rotation, while ψ_3 completes a full rotation every 2 samples (by convention, we consider completing a full rotation every 1 sample as doing no rotation at all, as that is a constant signal).

So a larger k does *not* mean a higher frequency.

2. DTFS and DFT, Its Fraternal Twin [Adapted from EE 120 Homework]

To represent finite-length or periodic discrete-time signals, engineers have traditionally used a complex exponential Fourier series expansion that is slightly different from the Discrete-Time Fourier Series (DTFS). The expansion of choice is called the Discrete Fourier Transform (DFT).

In this problem, you will discover the simple relationship between the DTFS and the DFT. Recall that the analysis and synthesis equations of the DTFS expansion of a periodic discrete-time signal $x : \mathbb{Z} \rightarrow \mathbb{C}$ are:

$$x[n] = \sum_{k=0}^{p-1} X_k e^{ik\omega_0 n} \quad \xleftrightarrow{\text{DTFS}} \quad X_k = \frac{1}{p} \sum_{n=0}^{p-1} x[n] e^{-ik\omega_0 n}$$

where $\omega_0 = 2\pi/p$ is the fundamental frequency of the signal.

- (a) Consider the set of basis signals $\{\boldsymbol{\phi}_0, \dots, \boldsymbol{\phi}_{p-1}\}$, where $\boldsymbol{\phi}_k = \frac{1}{p} \boldsymbol{\psi}_k$. Determine $\langle \boldsymbol{\phi}_k, \boldsymbol{\phi}_l \rangle$ for each of the cases $k = l$ and $k \neq l$.

Hint: What do you know about $\langle \boldsymbol{\psi}_k, \boldsymbol{\psi}_l \rangle$?

Answer:

$$\begin{aligned} \langle \boldsymbol{\phi}_k, \boldsymbol{\phi}_l \rangle &= \sum_{n \in \langle p \rangle} \phi_k[n] \phi_l^*[n] \\ &= \frac{1}{p^2} \sum_{n \in \langle p \rangle} \psi_k[n] \psi_l^*[n] \\ &= \frac{1}{p^2} \langle \boldsymbol{\psi}_k, \boldsymbol{\psi}_l \rangle \\ &= \frac{1}{p^2} (p \delta(k-l)) \\ &= \boxed{\frac{1}{p} \delta(k-l)} \end{aligned}$$

So, when $k = l$, we have $\delta(0) = 1$ and when they're different we have a non-zero input so the delta becomes 0.

(b) We wish to express a p -periodic signal x in terms of the basis functions in $\{\boldsymbol{\varphi}_0, \dots, \boldsymbol{\varphi}_{p-1}\}$, as follows:

$$x = \sum_{k \in \langle p \rangle} X'_k \boldsymbol{\varphi}_k.$$

Determine an expression for the coefficients X'_k in terms of projections of the signal x onto the basis functions $\boldsymbol{\varphi}_k$.

- i. What is the relationship between the DFT coefficients X'_k and the DTFS coefficients X_k ?
- ii. Write down the analysis and synthesis equations for the DFT.

Answer:

$$\begin{aligned} X'_k &= \frac{\langle x, \boldsymbol{\varphi}_k \rangle}{\langle \boldsymbol{\varphi}_k, \boldsymbol{\varphi}_k \rangle} \\ &= \frac{1}{p} \sum_{n \in \langle p \rangle} x[n] \boldsymbol{\varphi}_k^*[n] \\ &= \sum_{n \in \langle p \rangle} x[n] \psi_k^*[n] \\ &= \boxed{pX_k} \end{aligned}$$

Analysis:

$$\begin{aligned} X'_k &= pX_k \\ &= \boxed{\sum_{n \in \langle p \rangle} x[n] e^{-ik\omega_0 n}} \end{aligned}$$

Synthesis:

$$\begin{aligned} x[n] &= \sum_{k \in \langle p \rangle} X'_k \boldsymbol{\varphi}_k[n] \\ &= \boxed{\frac{1}{p} \sum_{k \in \langle p \rangle} X'_k e^{ik\omega_0 n}} \end{aligned}$$